

Module 4: Sustainability and its Assessment

10 January 2023

Assoc. Prof. Ir. Dr.-Ing. Mohamad Rizza Othman

Trainer Background

- Name : Mohamad Rizza Othman
- Position : Assoc. Prof. FTKKP & Deputy Director PHKK
- Education background
 - B. Eng. (Chem. Eng.) – UTM 2002
 - M. Eng. (Chem. Eng.) – UTM 2005
 - Dr.-Ing – TUB Germany 2011
- Professional qualifications
 - PEng or Ir. (BEM)
 - CEng (Engineering Council UK)
 - Member IChemE UK
 - Grad. Eng./Techn.
 - Aspen Plus Certified User
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Area of Expertise

- >15 years experience in Aspen Plus (Certified User), Dynamics, ASW, EDR and ANN.
- Taught UG/PG courses i.e., Process Control & Dynamics, Process Simulation & Advanced Process Simulation.
- Research area
 - Process modeling and simulation of various processes i.e., biodiesel, acetic acid, hydrogen, oleochemical, distillation, extraction, DWC, MSR, CO₂ absorption etc.
 - Safety analysis and sustainability
 - ANN
 - Optimization and multi criteria decision making
- Consultation projects
 1. Bio-MEG purification with PRSB
 2. Design and development of modular clinical waste incinerator (Radicare Sdn. Bhd.)
 3. Design and development of training rigs (Politeknik Pagoh)

Module Objectives

At the end of this module, you will be able to:

1. Describe the concept of sustainability, its roles, importance and applications in process industries.
2. Identify approaches for sustainable process design and assess sustainability performance in process industries.

Outline

10 January 2023

1430 – 1700

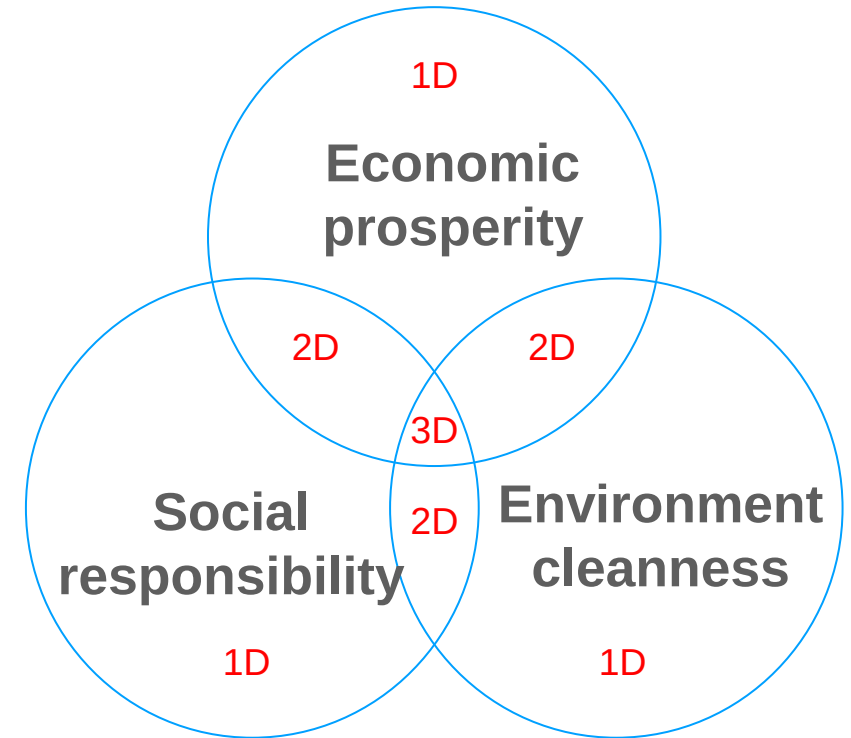
- Introduction to Sustainability
- Sustainability assessment and analysis
 - Economy
 - Environment
 - Social

Definition of Sustainability

General definition:-

*“... development must meet the **needs of the present** without **compromising** the ability of **future generations to meet their own needs**”.* The Brundtland Commission

- The development of sustainability increases and broaden the definition i.e., cleaner production, pollution prevention, pollution control, and minimization of resource usage, eco-design etc.
- The Triple Bottom Line (TBL) is commonly used to describe organizational sustainability based on 3 principles of sustainability; economic, environmental and society.



3 Dimensions of Sustainability

Sustainability Principles

Economy

- Economic principles composed of terms such as environmental accounting, eco-efficiency and ethical investments.

Environment

- Describe environmental or ecological performance to minimize the use of hazardous or toxic substances, resources and energy.
- Include terms i.e., renewable resources, resource minimization, source reduction, recycling, reuse, repair, regeneration, recovery, remanufacturing, purification, end-of-pipe, degradation etc.

Social

- Social Responsibility that refers to safe, respectful, liberal, equitable and equal human development, contributing to humanity and the environment.
- Health and safety refers to the working environment and includes responsibilities and standards.

Sustainability Development Goals (SDG)

- Decades of development started in June 1992, at the Earth Summit in Rio de Janeiro, Brazil,
- The 2030 Agenda for SD provides a shared blueprint for peace and prosperity for people and the planet, now and into the future.

“... ending poverty and other deprivations must go hand-in-hand with strategies that improve health and education, reduce inequality, and spur economic growth – all while tackling climate change and working to preserve our oceans and forests.”

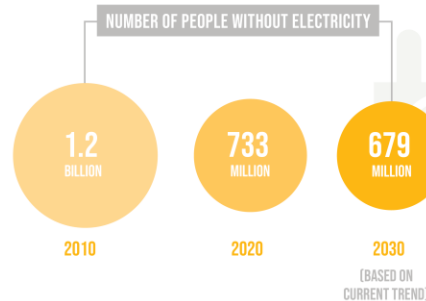




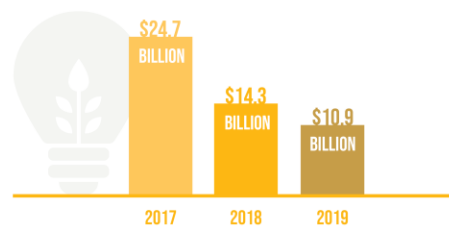
ENSURE ACCESS TO AFFORDABLE, RELIABLE, SUSTAINABLE AND MODERN ENERGY FOR ALL

IMPRESSIVE PROGRESS IN ELECTRIFICATION HAS SLOWED

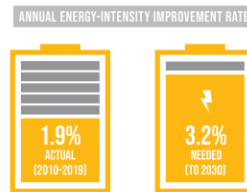
DUE TO THE CHALLENGE OF REACHING THOSE HARDEST TO REACH



INTERNATIONAL FINANCIAL FLOWS TO DEVELOPING COUNTRIES FOR RENEWABLES DECLINED FOR A SECOND YEAR IN A ROW

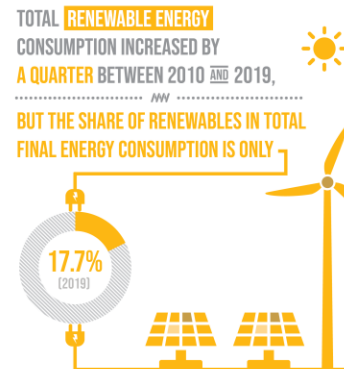


PROGRESS IN ENERGY EFFICIENCY NEEDS TO SPEED UP TO ACHIEVE GLOBAL CLIMATE GOALS



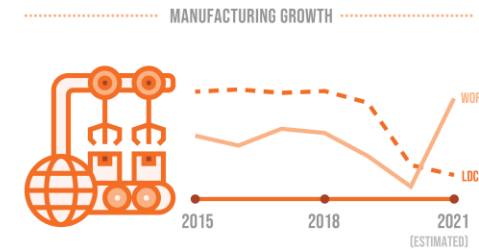
2.4 BILLION PEOPLE

STILL USE INEFFICIENT AND POLLUTING COOKING SYSTEMS (2020)

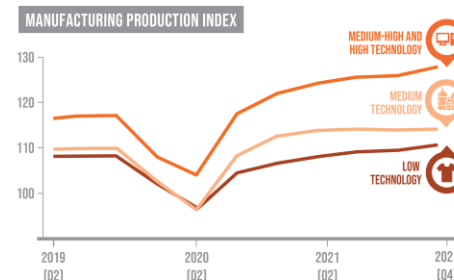


BUILD RESILIENT INFRASTRUCTURE, PROMOTE INCLUSIVE AND SUSTAINABLE INDUSTRIALIZATION AND FOSTER INNOVATION

GLOBAL MANUFACTURING HAS REBOUNDED FROM THE PANDEMIC BUT LDCs ARE LEFT BEHIND



HIGHER-TECHNOLOGY INDUSTRIES ARE FAR MORE RESILIENT IN CRISES THAN THEIR LOWER-TECH COUNTERPARTS

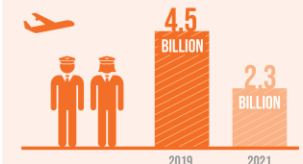


SMALL-SCALE INDUSTRIES



ONLY 1 IN 3 SMALL MANUFACTURERS ARE BENEFITING FROM A LOAN OR LINE OF CREDIT (2020-2021)

PASSENGER AIRLINE INDUSTRY IS STILL STRUGGLING TO RECOUP CATASTROPHIC LOSSES



2.3 BILLION PASSENGERS IN 2021, COMPARED WITH 4.5 BILLION IN 2019

1 IN 3 MANUFACTURING JOBS ARE NEGATIVELY IMPACTED BY THE CRISIS



ENSURE SUSTAINABLE CONSUMPTION AND PRODUCTION PATTERNS

UNSUSTAINABLE PATTERNS

OF CONSUMPTION AND PRODUCTION ARE ROOT CAUSE OF

TRIPLE PLANETARY CRISES



CLIMATE CHANGE



BIODIVERSITY LOSS



POLLUTION

TOO MUCH FOOD IS BEING LOST OR WASTED IN EVERY COUNTRY EVERY DAY



HARVESTING



TRANSPORT



STORAGE



PROCESSING

13.3% OF THE WORLD'S FOOD IS LOST AFTER HARVESTING AND BEFORE REACHING RETAIL MARKETS



HOUSE



GROCERY STORE



HOUSEHOLD



RESTAURANT

17% OF TOTAL FOOD IS WASTED AT THE CONSUMER LEVEL

OUR RELIANCE ON NATURAL RESOURCES IS INCREASING

RISING OVER 65% GLOBALLY FROM 2000 TO 2019



VAST MAJORITY OF THE WORLD'S ELECTRONIC WASTE IS NOT BEING SAFELY MANAGED

E-WASTE COLLECTION RATES (2019)



LATIN AMERICA AND THE CARIBBEAN



SUB-SAHARAN AFRICA



EUROPE AND NORTHERN AMERICA



GLOBAL AVERAGE

TAKE URGENT ACTION TO COMBAT CLIMATE CHANGE AND ITS IMPACTS

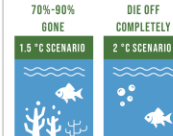


CLIMATE CHANGE

IS HUMANITY'S "CODE RED" WARNING

OUR WINDOW TO AVOID CLIMATE CATASTROPHE IS CLOSING RAPIDLY

DIFFERENT TEMPERATURE SCENARIOS FOR CORAL REEFS



CORAL REEFS

SEA LEVEL WILL RISE 30-60 CM BY 2100



SEA LEVEL RISE

DROUGHT ESTIMATED TO DISPLACE 700 MILLION PEOPLE BY 2030



DROUGHTS

MEDIUM- TO LARGE-SCALE DISASTERS WILL INCREASE 40% FROM 2015 TO 2030



DISASTERS

ENERGY-RELATED CO₂ EMISSIONS INCREASED 6% IN 2021

REACHING HIGHEST LEVEL EVER

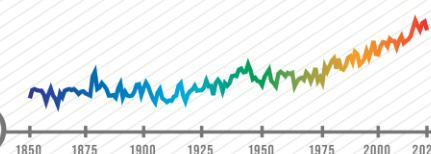


CLIMATE FINANCE FALLS SHORT OF \$100 BILLION YEARLY COMMITMENT

DEVELOPED COUNTRIES PROVIDED \$79.6 BILLION IN CLIMATE FINANCE IN 2019

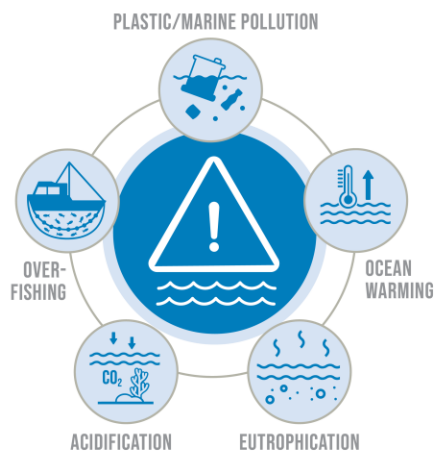


RIISING GLOBAL TEMPERATURES CONTINUE UNABATED, LEADING TO MORE EXTREME WEATHER



CONSERVE AND SUSTAINABLY USE THE OCEANS, SEA AND MARINE RESOURCES FOR SUSTAINABLE DEVELOPMENT

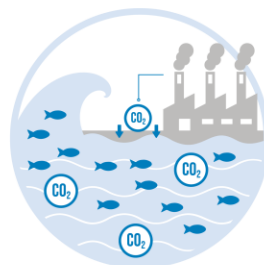
OUR OCEAN THE PLANET'S LARGEST ECOSYSTEM IS ENDANGERED



PLASTIC POLLUTION
IS CHOKING THE OCEAN

17+ MILLION METRIC TONS
OF PLASTIC ENTERED
THE OCEAN IN 2021

PROJECTED TO DOUBLE OR
TRIPLE BY 2040



INCREASING ACIDIFICATION
IS THREATENING MARINE LIFE AND
LIMITING THE OCEAN'S CAPACITY
TO MODERATE CLIMATE CHANGE

THE OCEAN ABSORBS AROUND 1/4
OF GLOBAL ANNUAL CO₂ EMISSIONS



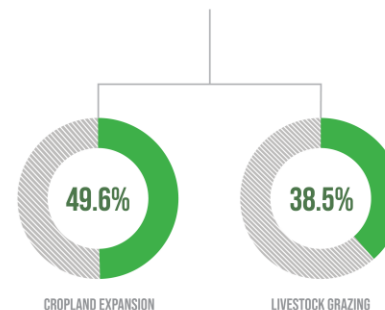
90% OF THE
WORLD'S FISHERS
ARE EMPLOYED IN
SMALL-SCALE FISHERIES
WHO NEED ACCELERATED
SUPPORT DUE TO THE
PANDEMIC

PROTECT, RESTORE AND PROMOTE SUSTAINABLE USE OF TERRESTRIAL ECOSYSTEMS, SUSTAINABLY MANAGE FORESTS, COMBAT DESERTIFICATION, AND HALT AND REVERSE LAND DEGRADATION AND HALT BIODIVERSITY LOSS

10 MILLION

HECTARES OF FOREST ARE DESTROYED EVERY YEAR

ALMOST 90% OF GLOBAL DEFORESTATION
IS DUE TO AGRICULTURAL EXPANSION



133 PARTIES HAVE RATIFIED
THE NAGOYA PROTOCOL,
WHICH ADDRESSES ACCESS TO
GENETIC RESOURCES
AND THEIR FAIR AND EQUITABLE USE

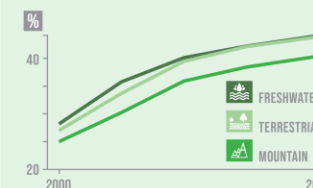
BIODIVERSITY
IS LARGELY NEGLECTED
IN COVID-19 RECOVERY SPENDING



AROUND 40,000 SPECIES
ARE DOCUMENTED
TO BE AT RISK
OF EXTINCTION
OVER THE
COMING
DECADES



NEARLY HALF OF FRESHWATER, TERRESTRIAL
AND MOUNTAIN KEY BIODIVERSITY AREAS
ARE PROTECTED



Sustainability in Process Industry

- Most chemical, oil and gas companies aim towards achieving cost-efficient and environmentally friendly operations.
- The negative environmental impact of their activities has led to an unfavorable perception.
- The chemical, oil and gas industry are one of the largest emitters of GHG i.e., CO₂ and CH₄.
- Effect of chemicals used during various chemical, oil and gas production stages i.e., upstream operation on air, land, oceans, marine life etc.

Example of sustainability efforts and activities in the process industry.

What can the chemical industry do to maintain or enhance the 'stock' of the following resources, or 'capitals'	As a business, accountable to shareholders	As a provider of products and services	As a significant member of the community where it operates
NATURAL The resources and services provided by natural world	- Eliminate the use of non-renewable energy - Use energy efficiently - Use water efficiently - Use raw materials efficiently - Audit supply chain performance - Eliminate waste - Operate to a consistently high global standard of environmental and social performance - Increase resource productivity	- Develop products that reduce energy use of customers - Develop products that copy natural processes - Improve biodegradability - Remove toxicity - Eliminate persistent compounds - Design products for reuse and recycling - Service models replace products	- Protect and enhance biodiversity - Commit to long-term carbon neutrality - Eliminate noise and odour - Eliminate visual intrusion - Eliminate negative impacts of local air quality
HUMAN The energy, motivation, capacity for relationships, and intelligence of individuals	- Implement (continuous) employee training and development schemes - Develop leadership - Implement diversity and inclusiveness programmes - Protect health and safety of employees - Provide opportunities for personal growth	- Eliminate negative health and safety impacts of products - Design products that meet human needs and enhance quality of life	- Foster local employment - Develop employee volunteering programmes - Engage in dialogue - Protect human rights
SOCIAL The social groupings that add value to individuals (e.g. families, communities, parliaments, universities)	- Encourage local procurement - Ensure that employees understand company vision, policies and programmes. - Support and encourage progressive regulation	- Communicate information related to product performance, risks and appropriate use. - Assist consumers to understand the impact of their actions and consumption patterns.	- Sustainability performance is openly and accurately communicated - Develop partnerships with the local community - Involved in education programmes - Community investment programme aligned to sustainable development
MANUFACTURED The "stuff" that already in terms of infrastructure n terms of the tools, machines, roads, buildings in which we live and work, and so on.	- Maximise process efficiency - Reduce volumes of throughput (energy, raw materials etc) for each unit of output) - Audit supply chain performance	Infrastructure encourages product reuse and recycling	- Provide communities with appropriate access to and use of physical assets. For example, community groups provided with use of office space etc outside of normal working hours. - Continued investment in the maintenance and development of infrastructure that reduces risk of negative impact on the community.
FINANCIAL The money, stocks etc. that enable us to put a value on, and buy and sell, the above resources and ways that value can more accurately represent the real 'cost' of using them.	- Makes acceptable financial returns - Account for total cost of activities encompassing both intangibles, risk and externalities - Shift the focus of management compensation from short term financial performance to include areas of sustainability performance	- Total cost accounting is reflected along value chain - Create economically viable products - Research and development priorities aligned to sustainability objectives	- Contribute to local economies through appropriate taxation in all areas of operation. - Eliminate corruption - Philanthropy aligned to strategic vision of company - Systematically avoid any "legacy effects" associated with operations and products

An Evolving Sustainability Agenda

- **Objective:** from regulatory compliance to a broader social “license to operate”
- **Focus:** from health, safety and environment and contribution to society to reduction of emissions and support for the energy transition
- **Stakeholders:** from shareholders, regulators, employees and local communities to customers, investors, non-Governmental institutions and society in general
- **Reporting:** from voluntary and ad-hoc to standard, regular and transparent
- **Governance:** from embedded within HSE or line functions to Board level oversight and accountability

Sustainability Indicators

- Screening tools or indicators are useful to assess/measure product or process sustainability performance.
- Currently there is no standard method for measuring sustainability.
- Most indicator works independently.
- Assessment tools were developed that integrate the individual indicators to assess/measure sustainability.
- Indicators can be categorized as hard (quantitative) and soft (qualitative).

Hard (Quantitative)	Soft (Qualitative)
• Measurable & quantifiable. • Represented by formulars/equations and resulted numerical values. • Example : • Net present value (NPV) • Life cycle analysis (LCA) • WAR algorithm • Fault tree analysis (FTA) • IChemE Metrics • Emergy • etc.	• Based on human knowledge & experience. • Important in process assessment and decision making. • Example : • What if? Analysis • Cause-consequences diagram • Industrial hygiene reviews • etc.

Sustainability Assessment Tools for Process Industries

Global
Reporting
Initiative (GRI)

AIChE
Sustainability
Index

IChEmE
Sustainability
Metrics

12 Principles of
Green
Chemistry

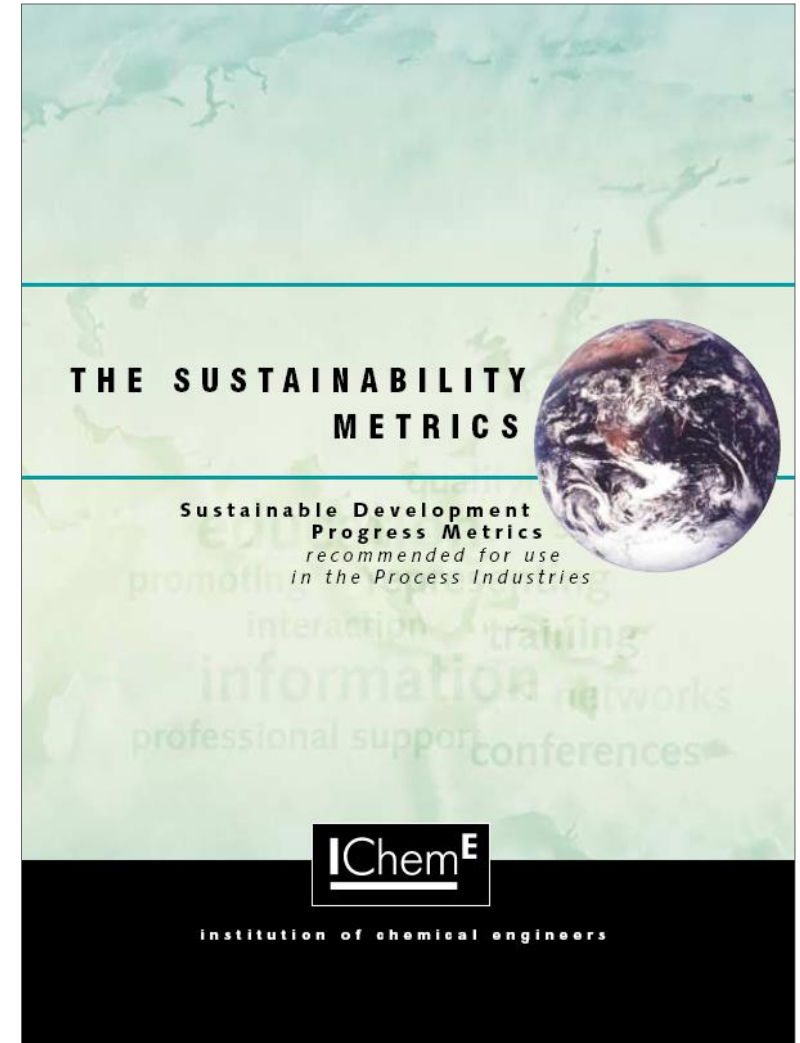
REACH (The
Registration,
Evaluation,
Authorization
and Restriction
of Chemicals)

TABLE 1: Main principles/evaluation guidelines of reviewed assessment methods [5, 12–16].

	GRI	AIChE	IChEmE	Green Chemistry
Environmental performance	Materials Energy Water Biodiversity Emissions Effluents and waste Products and services Compliance Transport Suppliers	(i) Resource use (a) Energy (b) Materials (c) Renewables (d) Water GHG emissions Waste , wastewater Compliance management Value chain management	Resource usage (i) Energy (ii) Material (iii) Water (iv) Land Emissions, effluents, and waste	Prevent waste Use renewable feedstock Avoid chemical derivatives Catalysts Product degradability
Economic performance	Economic performance Market presence Procurement practices	(i) Sustainability innovation (ii) Strategic commitment to sustainability	Profit , value, tax (i) Investments	Maximise atom economy Increase energy efficiency
Social performance	Labor practices (i) Employment (ii) Health and safety (iii) Innovation and knowledge potential (iv) Diversity and equality society (i) Acceptability and social dialogue Human rights	Social responsibility (i) Stakeholder partnership (ii) Social investment (iii) Image in the community Product stewardship (i) assurance system (ii) risk communication (iii) legal proceedings	Workplace (i) Employment (ii) Health and safety society	Less hazardous chemical syntheses Safer chemicals, products, solvents, and reactions Accident prevention and real time analysis

IChemE Sustainability Metrics

- Develop by IChemE to help companies to set targets and develop internal standards and to monitor their sustainability progress in time.
- Environmental measures:-
 - How much resource are use i.e., energy, material, water.
 - Impacts to atmospheric, aquatic, and land caused by emissions, effluents, and waste.
- Economic measures:-
 - Profit gained
 - Value added
 - Taxes paid
 - Investments made
- Social measures:-
 - The employment situation
 - Health and safety at work
 - impacts to society



Remarks

- No standard method and not bounded to regulatory commitment.
- Multiple indicators tracks each individual improvement more efficiently compared to only a single indicator.
- The assessment objectives affect the boundaries and selection of indicators.
- As proses development progresses complex indicators are applicable.
- Indicators for early process design assessment should use simple algorithm and less data extensive but still maintain its relevancy compared to already operated plant.

Our early work on Sustainable Process Design & Decision Making

7870

Ind. Eng. Chem. Res. **2010**, 49, 7870–7881

A Modular Approach to Sustainability Assessment and Decision Support in Chemical Process Design

Mohamad R. Othman,^{*,†,‡} Jens-Uwe Repke,[†] Günter Wozny,[†] and Yinlun Huang[§]

Chair of Process Dynamics and Operation, Berlin Institute of Technology, Germany, Department of Chemical Engineering & Material Science, Wayne State University, Detroit, Michigan, and Faculty of Chemical Engineering & Natural Resources, Universiti Malaysia Pahang, Pahang Darul Makmur, Malaysia

In chemical and allied industries, process design sustainability has gained public concern in academia, industry, government agencies, and social groups. Over the past decade, a variety of sustainability indicators have been introduced, but with various challenges in application. It becomes clear that the industries need urgently practical tools for conducting systematic sustainability assessment on existing processes and/or new designs and, further, for helping derive the most desirable design decisions. This paper presents a systematic, general approach for sustainability assessment and design selection through integrating hard (quantitative) economic and environmental indicators along with soft (qualitative) indicators for social criteria into design activities. The approach contains four modules: a process simulator module, an equipment and inventory acquisition module, a sustainability assessment module, and a decision support module. The modules fully utilize and extend the capabilities of the process simulator Aspen Plus, Aspen Simulation Workbook, and a spreadsheet, where case model development, data acquisition and analysis, team contribution assessment, and decision

NPV & DCFRR

- Pintarič and Kravanja (2006) suggested net present value (NPV) and discounted cash flow rate of return (DCFRR) as economic indicators as it favored by entrepreneurs.
- NPV & DCFRR reflect a comprehensive economic assessment of the overall project's economic life cycle.
- NPV provide profitability measurement whereas DCFRR reflect the highest, after-tax interest or discount rate at which the project can just break even.

$$NPV = \sum_{m=1}^n \frac{C_{A,m}}{(1+r)^m} - C_{TCI}$$

C_A = total annual income cash flow after the base year for year m

r = interest rate (%)

n = project life time after the base year

C_{TCI} = total capital investment

DCFRR = r or r_{ROR} when $NPV = 0$.

Higher r_{ROR} is more preferred.

$$0 = \sum_{m=1}^n \frac{C_{A,m}}{(1+r)^m} - C_{TCI}$$

Example

- Calculate NPV when $r = 8\%$

$$NPV = \sum_{m=1}^n \frac{C_{A,m}}{(1+r)^m} - C_{TCI}$$

- Calculate DCFRR

$$0 = \sum_{m=1}^n \frac{C_{A,m}}{(1+r)^m} - C_{TCI}$$

- Which is better?

Period	Cash flow 1, \$	Cash flow 2, \$
0	-1000	-1000
1	475	400
2	400	425
3	330	390
4	270	150
5	200	250
NPV	379.3	324.7
DCFRR	23.9%	21.4%

Reaction Cost

Reaction Path	Overall reaction
1. Hydrochlorination of acetylene	$\text{C}_2\text{H}_2 + \text{HCl} \rightarrow \text{C}_2\text{H}_3\text{Cl}$
2. Thermal cracking of 1,2-dichloroethane from chlorination of ethylene	$\text{C}_2\text{H}_4 + \text{Cl}_2 \rightarrow \text{C}_2\text{H}_3\text{Cl} + \text{HCl}$
3. Thermal cracking of 1,2-dichloroethane from oxychlorination of ethylene	$\text{C}_2\text{H}_4 + \text{HCl} + \frac{1}{2}\text{O}_2 \rightarrow \text{C}_2\text{H}_3\text{Cl} + \text{H}_2\text{O}$

- Example: Vinyl chloride production.
- Assess the cost of the reaction pathway at molecular level.

Chemical	Cost (cents/lb)
Ethylene	30
Acetylene	80
Chlorine	18
Vinyl chloride	35
Hydrogen chloride	25
Water	0
Oxygen (air)	0

Calculation example

Pathway	C2H2	Cl2	C2H3Cl	HCl
Stoic. coeff., lb-mole	-1	-1	1	1
MW	28.05	70.91	62.50	36.46
Mass, lb	28.05	70.91	62.50	36.46
lb/lb VC	0.449	1.134	1	0.583
Cents/lb	30	18	35	25
Cost, Cents/lb VC	-13.47	-20.412	35	14.575
Gross profit, Cents/lb VC	- (13.47 + 20.412) + (35 + 14.575) = 15.693			

Exercise

Calculate the gross profit for each pathway and compare the results.

Reaction Path	Overall reaction	Gross profit, Cents/lb VC
1. Hydrochlorination of acetylene	$\text{C}_2\text{H}_2 + \text{HCl} \rightarrow \text{C}_2\text{H}_3\text{Cl}$	
2. Thermal cracking of 1,2-dichloroethane from chlorination of ethylene	$\text{C}_2\text{H}_4 + \text{Cl}_2 \rightarrow \text{C}_2\text{H}_3\text{Cl} + \text{HCl}$	
3. Thermal cracking of 1,2-dichloroethane from oxychlorination of ethylene	$\text{C}_2\text{H}_4 + \text{HCl} + \frac{1}{2}\text{O}_2 \rightarrow \text{C}_2\text{H}_3\text{Cl} + \text{H}_2\text{O}$	

Environmental Analysis

- IChemE Sustainability Metrics
- Waste Reduction Algorithm (WAR) - Young and Cabezas (1999)
- Environmental Life Cycle Assessment of Products - Heijungs et al., 1992

$$I_{i,c} = \alpha_{i,c} \cdot M_c$$

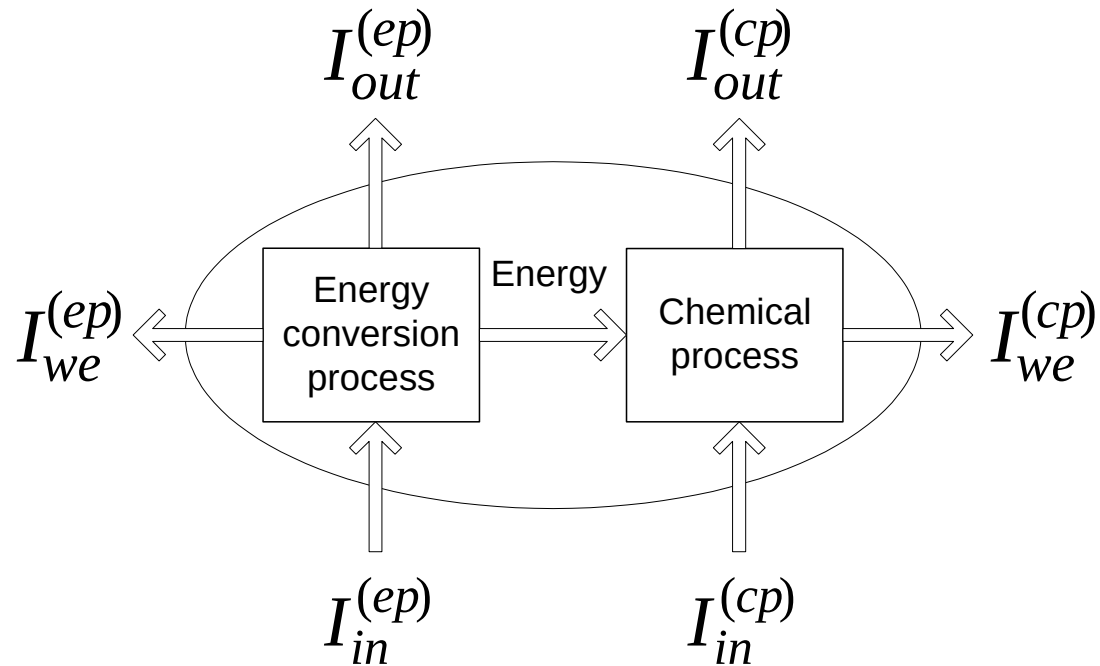
i = Impact category

c = component

α = Index or factor for impact category

M = mass of component, c

WAR (Waste Reduction) Algorithm



- Introduced by Young and Cabezas (1999) for assessing environmental impact at early process design stage.
- Simple to use algorithm and easy to find data.

$$I_{in}^{cp} + \cancel{I_{in}^{ep}} - I_{out}^{cp} - I_{out}^{ep} - \cancel{I_{we}^{cp}} - \cancel{I_{we}^{ep}} + I_{gen}^t = 0$$

Energy source is assumed in solid

Can be neglected due to small amount

Impact category

Category	Definition	WAR	IChemE	Heijungs et al.
Global warming potential (GWP)	Relative to CO ₂ , with time horizons of 20, 100 and 500 years	√	√	√
Ozone depletion potential (ODP)	The potency factor for this category is based on the potential to deplete ozone in the upper atmosphere relative to chlorofluorocarbon – 11	√	√	√
Acidification potential (AP)	The potential of certain released gases to form acid rain and acids to water is the potency factor for atmospheric acidification. Atmospheric acidification (IChemE).	√	√	√
Photochemical oxidation (or smog formation) potential (PCOP)	Potency factors for this category are obtained from the potential of substances to create ozone photochemically. Photochemical ozone creation potential (POCP) relative to ethylene (Heijungs et al.)	√	√	√
Human toxicity potential by ingestion (HTPI). Terrestrial toxicity potential (TTP)	LD ₅₀ (WAR). Occupational exposure limits (OEL). Benzene as reference (IChemE). Human lexicological classification factor for the air (HCA), human lexicological classification factor for water (HCW) and human toxicological classification factor for the soil (HCS) (Heijungs et al.).	√	√	√
Aquatic toxicity potential (ATP)	LC ₅₀	√		
Human toxicity potential by inhalation/dermal exposure (HTPE)	TWA-TLV	√		
Aquatic Acidification	The potency factor is the mass of hydrogen ion released by unit mass of acid. Measured pH values may also be used.		√	
Aquatic Oxygen Demand	The Stoichiometric Oxygen Demand (StOD) has been chosen as the potency factor. It represents the maximum potential of emissions to water to remove dissolved oxygen that would otherwise support fish and other aquatic life.		√	
Ecotoxicity to Aquatic Life	The potency factor is equal to the reciprocal of the Environmental Quality Standard (EQS) divided by the reciprocal of the EQS of copper.		√	√
Eutrophication	Eutrophication is defined as the potential for over-fertilisation of water and soil, which can result in increased growth of biomass		√	
Nutrification	Nutrification potential (NP) relative to PO ₄ , based on the average composition of biomass CiofiHaoOioNieP, relative to phosphate			√
Odour	Odour threshold value in air (OTV) in mg-m ⁻³			√

Energy Env. Impact

- Energy source for utilities include direct energy (electricity) and indirect energy (steam).
- Summing all the energy requirements of the system i.e. compressors, reboilers, heat exchangers, cooling and reboiler pumps, refrigeration units, turbines, etc.
- Emissions calculations for utilities are calculated based on an emission factor standards i.e., European Commission Decision 2007/589/EC, US EPA Rule E9-5711 and EPA's 2020 update.

Emission Factor for the Coal-Fired Power Plant (US EPA, 1995)

Gas pollutants	EF, kg/kWh
SO _x	0.00272
NO _x (NO ₂ ,NO)	0.00181
CO ₂	0.3719
HCl	9.0 x 10 ⁻⁵
Methane	0.4763
Mercury (Hg)	4.944 x 10 ⁻⁹

Stationary Combustion

Fuel Type	Heat Content (HHV)	CO ₂ Factor	CH ₄ Factor	N ₂ O Factor	CO ₂ Factor	CH ₄ Factor	N ₂ O Factor
	mmBtu per short ton	kg CO ₂ per mmBtu	g CH ₄ per mmBtu	g N ₂ O per mmBtu	kg CO ₂ per short ton	g CH ₄ per short ton	g N ₂ O per short ton
Coal and Coke							
Anthracite Coal	25.09	103.69	11	1.6	2,602	276	40
Bituminous Coal	24.93	93.28	11	1.6	2,325	274	40
Sub-bituminous Coal	17.25	97.17	11	1.6	1,676	190	28
Lignite Coal	14.21	97.72	11	1.6	1,389	156	23
Mixed (Commercial Sector)	21.39	94.27	11	1.6	2,016	235	34
Mixed (Electric Power Sector)	19.73	95.52	11	1.6	1,885	217	32
Mixed (Industrial Coking)	26.28	93.90	11	1.6	2,468	289	42
Mixed (Industrial Sector)	22.35	94.67	11	1.6	2,116	246	36
Coal Coke	24.80	113.67	11	1.6	2,819	273	40
Other Fuels - Solid							
Municipal Solid Waste	9.95	90.70	32	4.2	902	318	42
Petroleum Coke (Solid)	30.00	102.41	32	4.2	3,072	960	126
Plastics	38.00	75.00	32	4.2	2,850	1,216	160
Tires	28.00	85.97	32	4.2	2,407	896	118
Biomass Fuels - Solid							
Agricultural Byproducts	8.25	118.17	32	4.2	975	264	35
Peat	8.00	111.84	32	4.2	895	256	34
Solid Byproducts	10.39	105.51	32	4.2	1,096	332	44
Wood and Wood Residuals	17.48	93.80	7.2	3.6	1,640	126	63
	mmBtu per scf	kg CO ₂ per mmBtu	g CH ₄ per mmBtu	g N ₂ O per mmBtu	kg CO ₂ per scf	g CH ₄ per scf	g N ₂ O per scf
Natural Gas							
Natural Gas	0.001026	53.06	1.0	0.10	0.05444	0.00103	0.00010

<https://www.epa.gov/sites/production/files/2020-04/documents/ghg-emission-factors-hub.pdf>

*Short ton = 2,000 pounds. Commonly used in the US. Also known as ton.

$$I_{out}^{(t)} = I_{out}^{(cp)} + I_{out}^{(ep)}$$

Total rate of PEI output (TRO) - Identify an appropriate site for a plant (a plant with low should be located in an ecologically sensitive area).

$$\hat{I}_{out}^{(t)} = \frac{I_{out}^{(cp)} + I_{out}^{(ep)}}{\sum_p \dot{P}_p}$$

Total rate of PEI output/product (TOP) - measures the efficiency of material utilization by a specific process per unit mass of products; it decreases when the mass rate of PEI emitted decrease or/and when product mass flow rate increase.

$$I_{gen}^{(t)} = I_{out}^{(cp)} - I_{in}^{(cp)} + I_{out}^{(ep)}$$

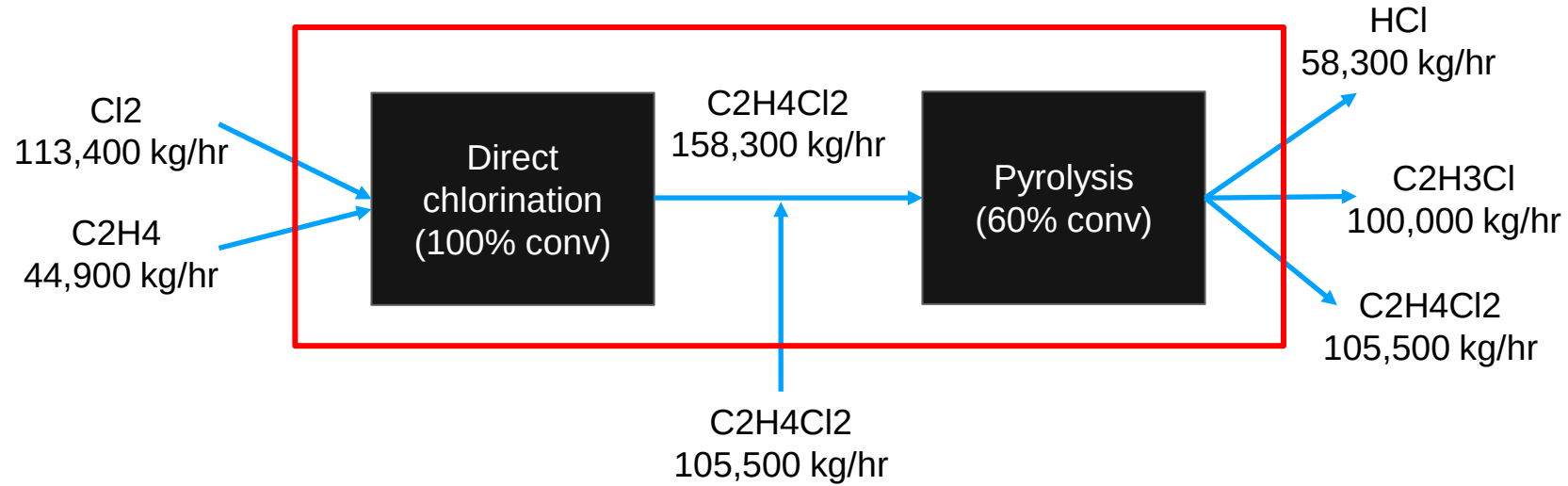
Total rate of PEI generated (TRG) - useful in comparing processes based on how fast they generate impact.

$$\hat{I}_{gen}^{(t)} = \frac{I_{out}^{(cp)} - I_{in}^{(cp)} + I_{out}^{(ep)}}{\sum_p \dot{P}_p}$$

Total rate of PEI generated/product (TGP) - is used for comparing processes and products based on the amount of new potential environmental impact generated in product manufacturing.

****Obviously, the lower the PEI, the more desirable the process.**

Illustrative example: VC production



Component	Input, kg/hr	Output, kg/hr
Chlorine, Cl ₂	113,400	
Ethylene, C ₂ H ₄	44,900	
1,2-dichloroethane, C ₂ H ₄ Cl ₂	105,500	105,500
Hydrogen chloride, HCl		58,300
VC, C ₂ H ₃ Cl		100,000
Energy		60,000 kW

Material Impact

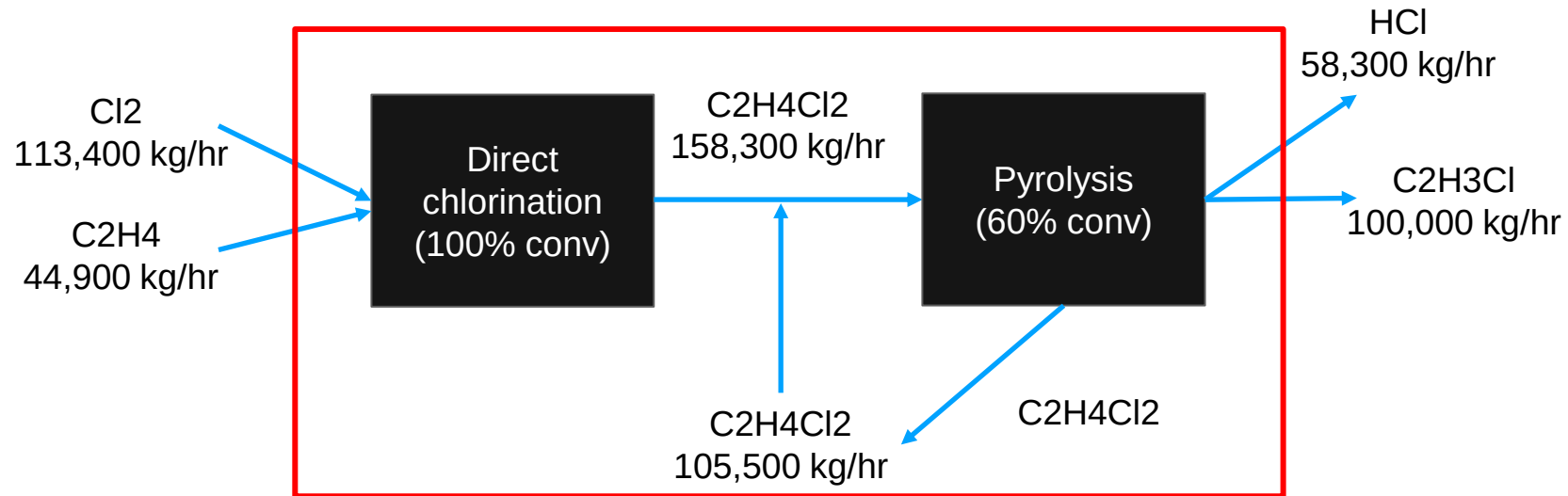


Component	PEI (GWP) Index	Input		Output	
		kg/h	PEI	kg/h	PEI
Chlorine, Cl2	1.1	113,400	124,740	-	-
Ethylene, C2H4	1.3	44,900	58,370	-	-
1,2 dichloroethane, C2H4Cl2	1.5	105,500	158,250	105,500	158,250
Hydrogen chloride, HCl	1.9	-	-	58,300	110,770
VC, C2H3Cl	1.7	-	-	100,000	170,000
TOTAL PEI (GWP)			341,360	439,020	

Energy Impact

Gas pollutants	EF, kg/kWh	PEI (GWP) Index	Input		Output	
			kWh	PEI	kg/h	PEI
Energy, kWh			60000	-	-	-
SOx	0.00272	1.2			163.2	195.84
NOx (NO2,NO)	0.00181	1.4			108.6	152.04
CO2	0.3719	1.6			22,314	35,702.4
HCl	9.0 x 10-5	1.1			5.4	5.94
Methane	0.4763	1.1			28,578	31,435.8
Mercury (Hg)	4.944 x 10-9	1.1			0.000297	0.000326
TOTAL PEI (GWP)					67,492.02	

After recycle



Component	Input, kg/hr	Output, kg/hr
Chlorine, Cl_2	113,400	
Ethylene, C_2H_4	44,900	
1,2-dichloroethane, $\text{C}_2\text{H}_4\text{Cl}_2$		
Hydrogen chloride, HCl		58,300
VC, $\text{C}_2\text{H}_3\text{Cl}$		100,000
Energy		65,000 kWh

Material Impact (Recycle)



Component	PEI (GWP) Index	Input		Output	
		kg/h	PEI	kg/h	PEI
Chlorine, Cl2	1.1	113,400	124,740	-	-
Ethylene, C2H4	1.3	44,900	58,370	-	-
1,2 dichloroethane, C2H4Cl2	1.5	-	-	-	-
Hydrogen chloride, HCl	1.9	-	-	58,300	110,770
VC, C2H3Cl	1.7	-	-	100,000	170,000
TOTAL PEI (GWP)			183,110	280,770	

Energy Impact (Recycle)

Gas pollutants	EF, kg/kWh	PEI (GWP) Index	Input		Output	
			kWh	PEI	kg/h	PEI
Energy, kWh			65,000	-	-	-
SOx	0.00272	1.2			176.8	212.16
NOx (NO2,NO)	0.00181	1.4			117.65	164.71
CO2	0.3719	1.6			24,173.5	38,677.6
HCl	9.0 x 10-5	1.1			5.85	6.435
Methane	0.4763	1.1			30,959.5	34,055.45
Mercury (Hg)	4.944 x 10-9	1.1			0.000321	0.000353
TOTAL PEI (GWP)					73,116.36	

Total Impact (Comparison)

	Before	After	Diff. %
Total mass PEI input	341,360	183,110	46%
Total mass PEI output	439,020	280,770	36%
Total energy PEI output	67,492.02	73,116.36	-8%
Total PEI Input (GWP)	341,360	183,110	46%
Total PEI Output (GWP) (Total mass PEI output + Total energy PEI output)	506,512.02	353,886.36	30%
Total PEI OUT/Mass product	5.1	3.5	31%
Total PEI (Out – In + Energy)	165,152	170,776	-3%

- Red color indicate favorable option.
- Each parameter has its own definition and should be consider based on the assessment objectives.
- Final decision making should consider other criteria i.e., economic potential and social aspects.

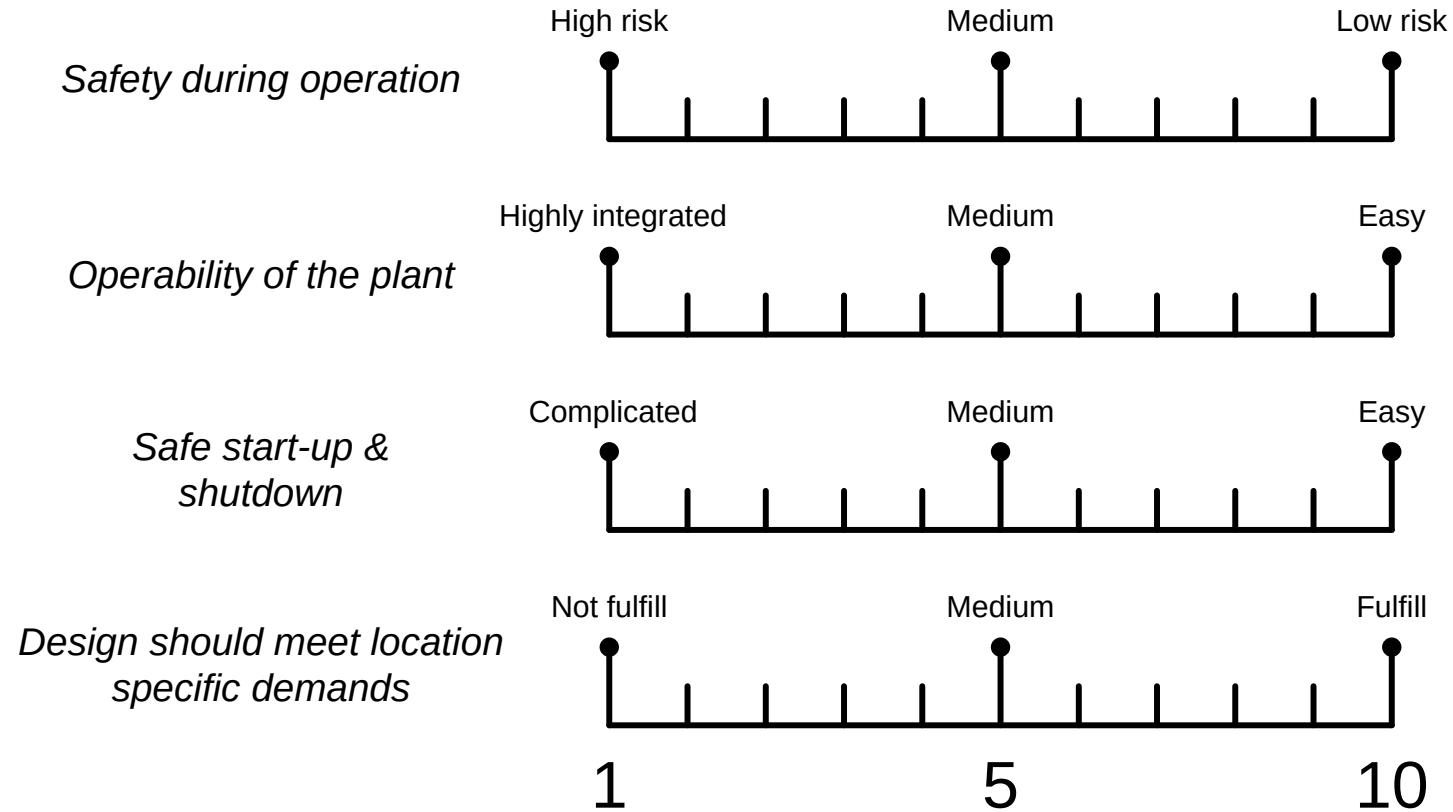
Social Indicators

- IChemE social indicators (quantitative)
- Reflects company's attitude to treatment of its own employees, suppliers, contractors and customers and also its impact on society at large.
- Irrelevant at early design stage esp. for back driven data.
- Herder and Weijnen (1998) define quality indicators for early design decision making.
- Utilize the heuristics knowledge.
- Provide a rapid assessment without the need for extensive data search.



- In our work (Othman et.al., 2010), 4 of them are adopted. As a guideline, each of the indicators is defined below. Although not very rigid, they should be useful to guide decision makers.

Indicators	Definition
Safety during operation	The condition or state of being safe; free from danger or hazard; exemption from hurt, injury, or loss. Evaluation of hazards and risks associated to, but not limited, chemical compounds, reactions, unit operations and equipments and operating conditions should include in the assessment.
Operability of the plant	The condition where the plant is able to operate feasibly. Assessment should consider the operation feasibility by workers and also control systems of the plant especially if tightly integrated and also in the presence of process variations and uncertainties.
Safe start-up and shutdown	Start-up means the act or process of setting into operation or motion while shut down means cease to operate or cause to cease operating. The degree of difficulties of the procedure depends on the system complexity and workers capability.
Design should meet location specific demands	Local demands may include technology transfer, employment, affect to other related industries, local regulations and policies, legal proceedings etc.



- Because of heuristics assessment a scaling system is proposed.
- Although not very specific, it acts as a general guideline to assess various types of chemical processes.